2.6 Carrier Transport

2.6.1 Carrier Drift
- Drift Current
- Mobility
- Resistivity
- Band Structure under Applied Field

2.6.2 Carrier Diffusion
- Diffusion Current

2.6.3 Total Current Equations

2.6.4 Einstein Relations

Pierret, Chapter 3.1-3.2, page 75-104

2.6.1 Carrier Drift

- **Drift = Charged particle motion in response to an applied electric field**
- Macroscopic Definition: Carriers of a given type (electrons or holes) move along at a constant velocity, the drift velocity, parallel or antiparallel to the applied electric field
- **Drift Current Densities** [A/cm^2]:

$$\boxed{\begin{aligned} \vec{J}_{p,drift} &= q\,p\,\vec{v}_{d,p} \\ \vec{J}_{n,drift} &= -q\,n\,\vec{v}_{d,n} \end{aligned}}$$

Carrier Drift

- For small electric fields, the drift velocity is proportional to the applied electric field with the mobility μ as proportionality factor:

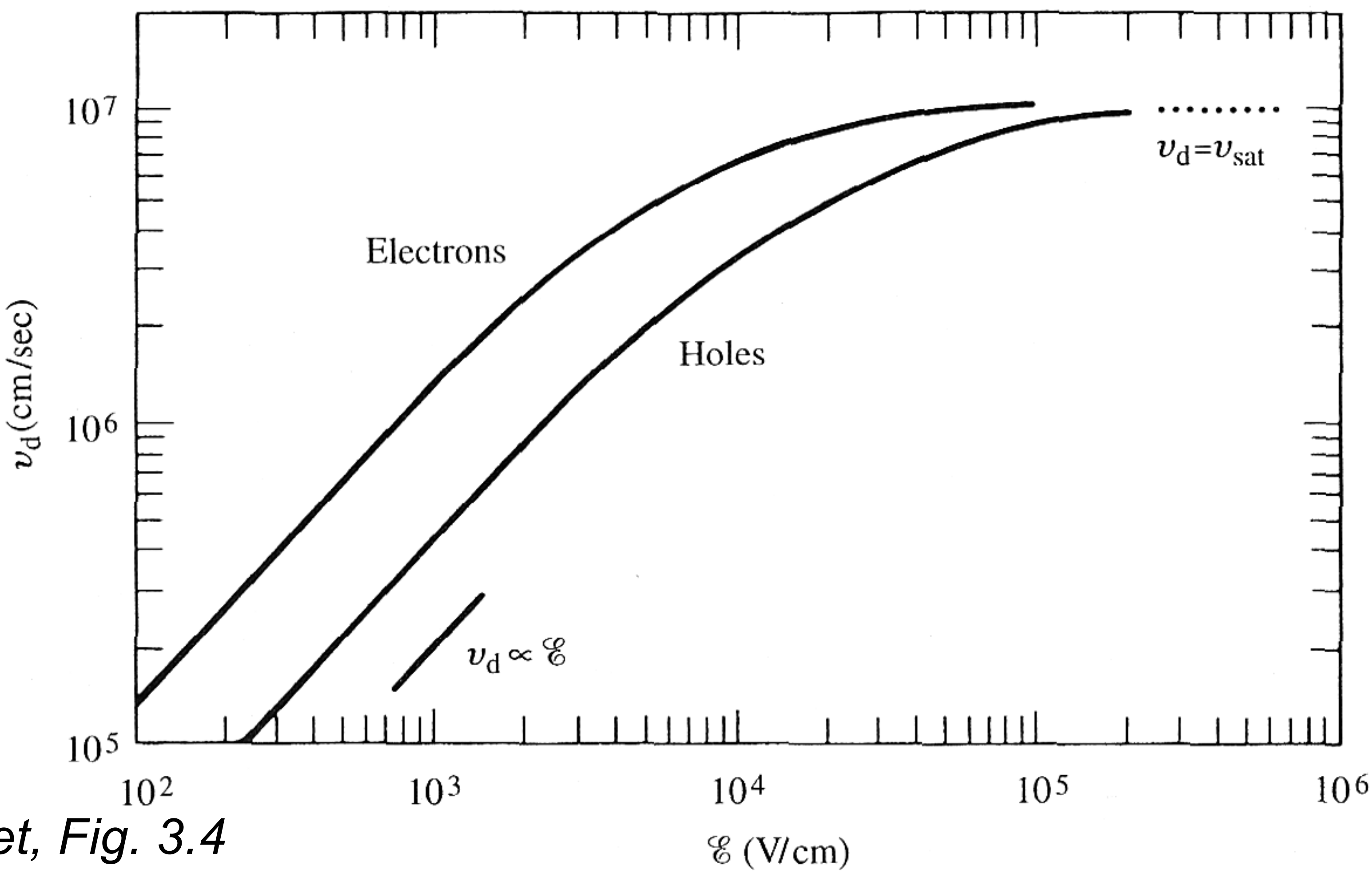
$$v_{d,p} = \mu_p \, \varepsilon$$

$$v_{d,n} = -\mu_n \, \varepsilon$$

- The mobility is the central parameter characterizing the carrier drift, resulting in the following current densities:

$$\boxed{\begin{aligned} \vec{J}_{p,drift} &= q\, p\, \mu_p \, \vec{\varepsilon} \\ \vec{J}_{n,drift} &= q\, n\, \mu_n \, \vec{\varepsilon} \\ \vec{J}_{drift} &= \vec{J}_n + \vec{J}_p = q\left[\mu_n n + \mu_p p\right]\vec{\varepsilon} \end{aligned}}$$

Drift Velocity in Undoped Si at Room Temperature



Pierret, Fig. 3.4

Carrier Mobility

- Definition of the carrier mobility [cm^2/Vs]:

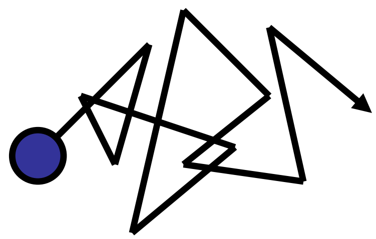
$$\boxed{\mu_p \equiv \frac{v_{d,p}}{\varepsilon} \qquad \mu_n \equiv -\frac{v_{d,n}}{\varepsilon}}$$

- Room-temperature mobility of Si and GaAs:

Silicon (low-doped)	$\mu_n \approx 1360$ cm^2/Vs $\mu_p \approx 460$ cm^2/Vs
Gallium Arsenide (low-doped)	$\mu_n \approx 8500$ cm^2/Vs $\mu_p \approx 400$ cm^2/Vs

Drift – Microscopic Picture

- Charged particles are accelerated by the electric field ($F = q\,\varepsilon$) and loose energy by collisions with impurities and the crystal lattice (**impurity scattering** and **lattice scattering**)



- The drift of a particle is superimposed on the random **thermal motion**

Scattering Mechanisms I

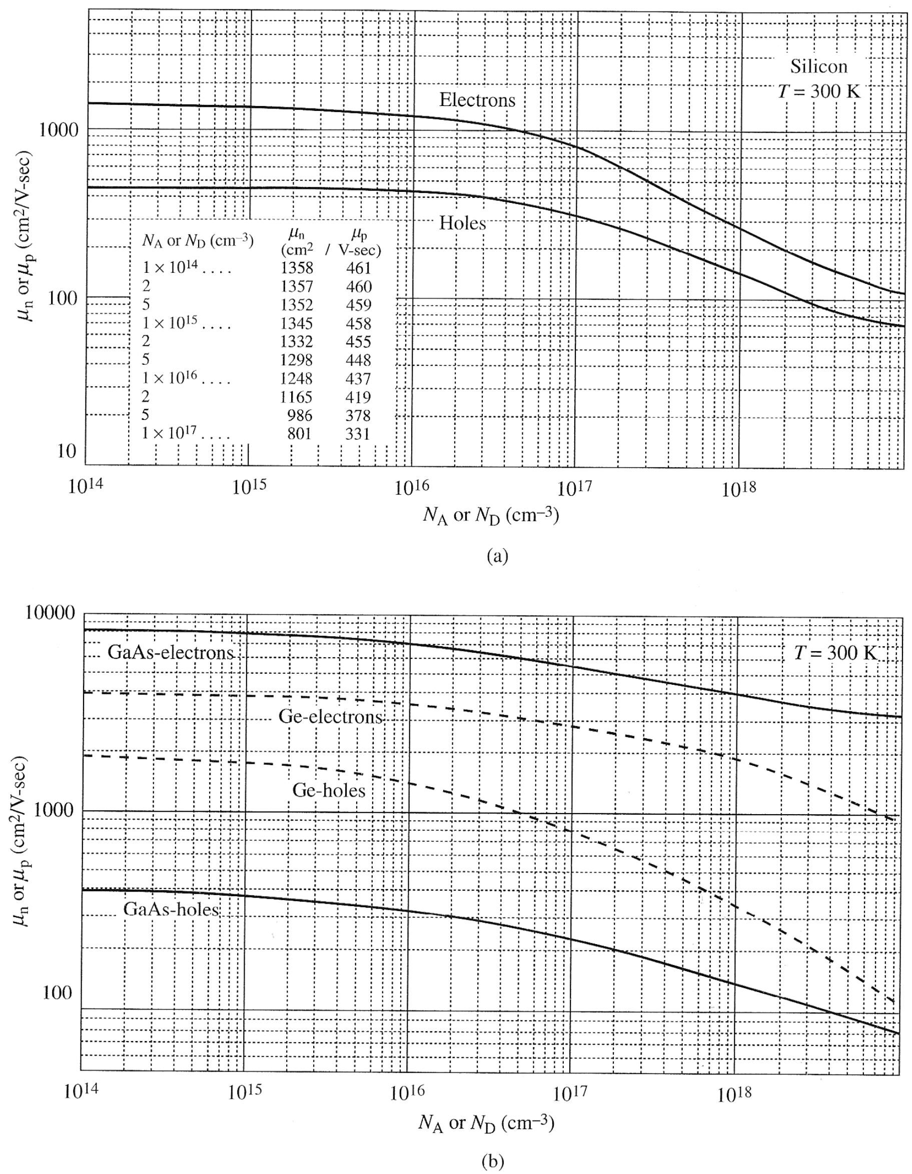
- (a) **Impurity** and (b) **lattice scattering** limit the carrier mobility

$$\mu = \frac{q\,\tau}{m^*}$$

mean free time between collisions

effective mass

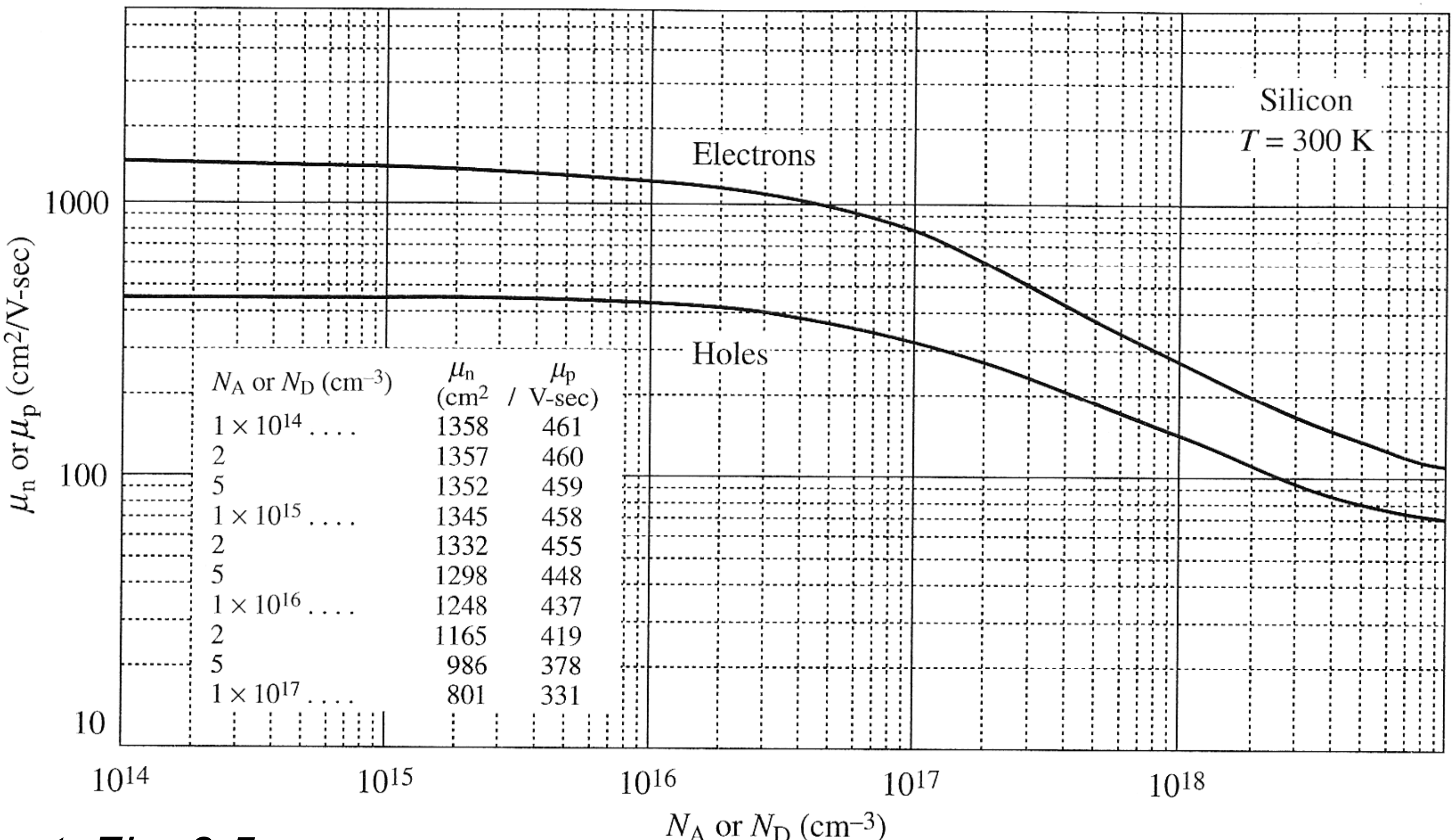
- The mobility decreases with increasing **total** doping concentration ($N_A + N_D$)



N_A or N_D (cm^{-3})	μ_n (cm²/	μ_p V-sec)
1×10^{14}	1358	461
2	1357	460
5	1352	459
1×10^{15}	1345	458
2	1332	455
5	1298	448
1×10^{16}	1248	437
2	1165	419
5	986	378
1×10^{17}	801	331

Pierret, Fig. 3.5

Room Temperature Mobility in Si as a Function of Doping Concentration



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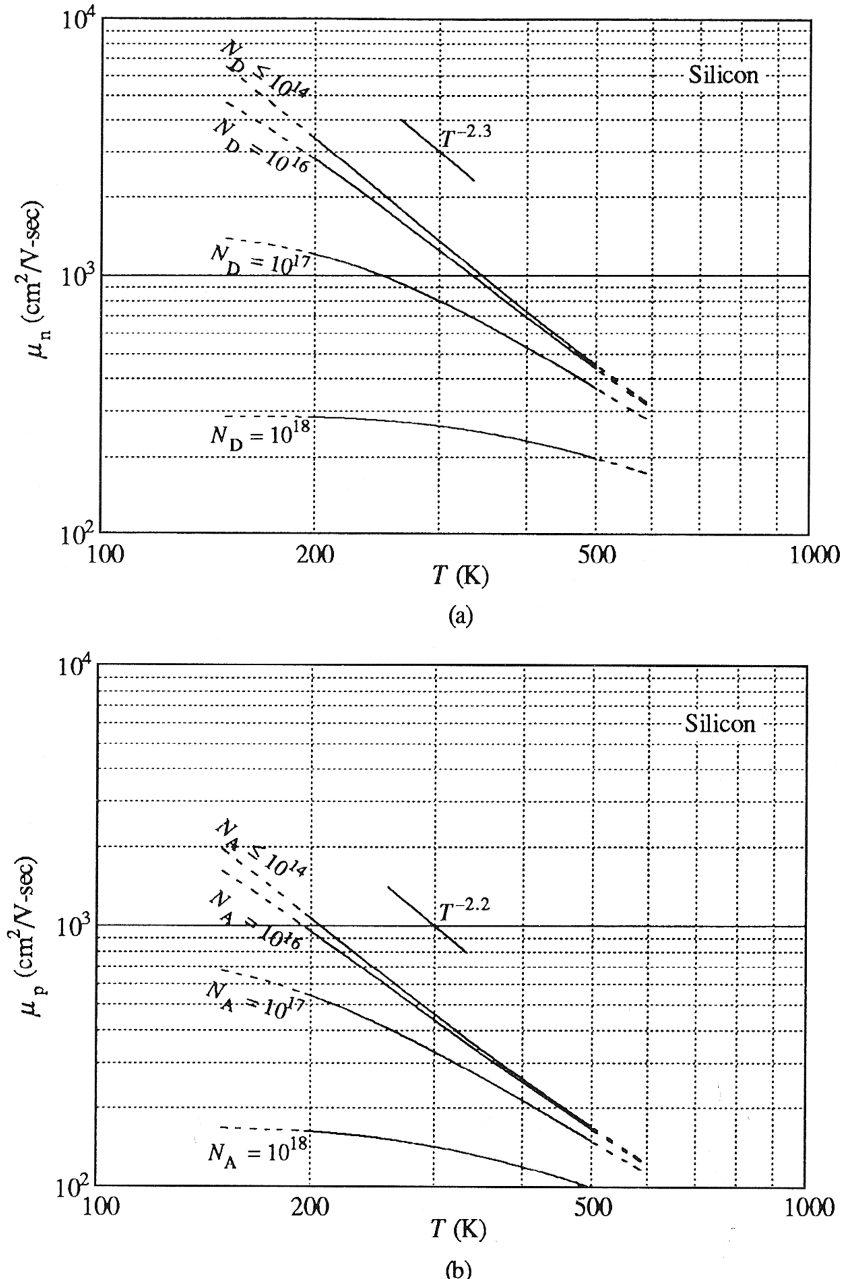
Pierret, Fig. 3.5a

Scattering Mechanisms II

T-dependence of the mobility:

- **Lattice Scattering:**
  in Si: $\mu \sim T^{-(2.2\ldots2.3)}$ (experiment)
  Reducing the temperature means less thermal lattice vibration, i.e. less interaction with carriers, i.e. higher mobility
- **Impurity Scattering:**
  in Si: $\mu \sim T^{+1.5}$ (theory)
  Dominant scattering mechanism at low temperatures; interaction is reduced at higher temperatures, i.e. higher thermal velocities, because carrier is less time in close proximity to impurity
- **$\mu = \mu(T, N_A+N_D)$**



Pierret, Fig. 3.7

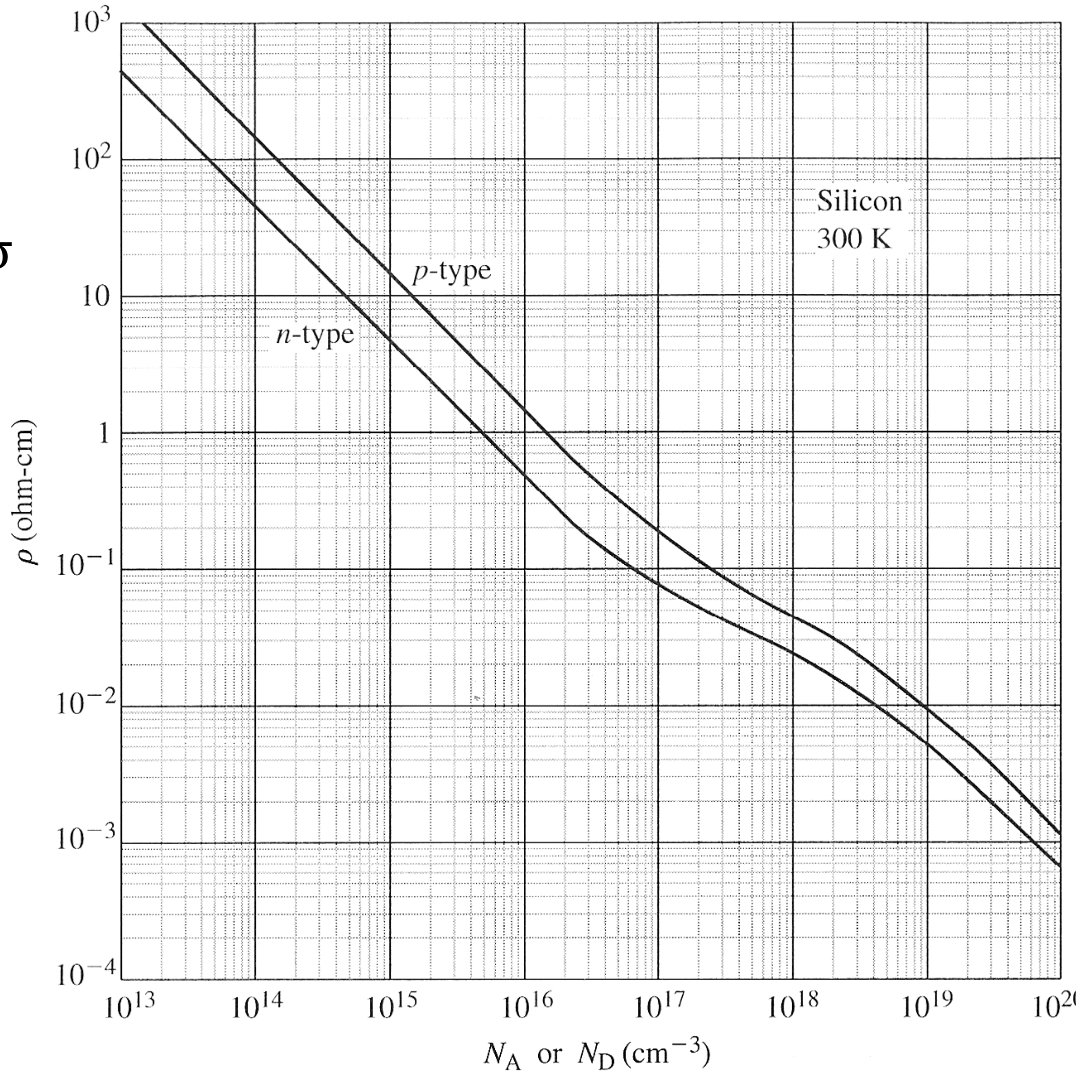
Resistivity

- Definition of conductivity σ and resistivity ρ [Ωcm] tensor:

$$\boxed{\vec{J} = \sigma\,\vec{\varepsilon} = \frac{1}{\rho}\,\vec{\varepsilon}}$$

- From the drift current, the resistivity is given by:

$$\boxed{\rho = \frac{1}{q\left(\mu_n n + \mu_p p\right)}}$$



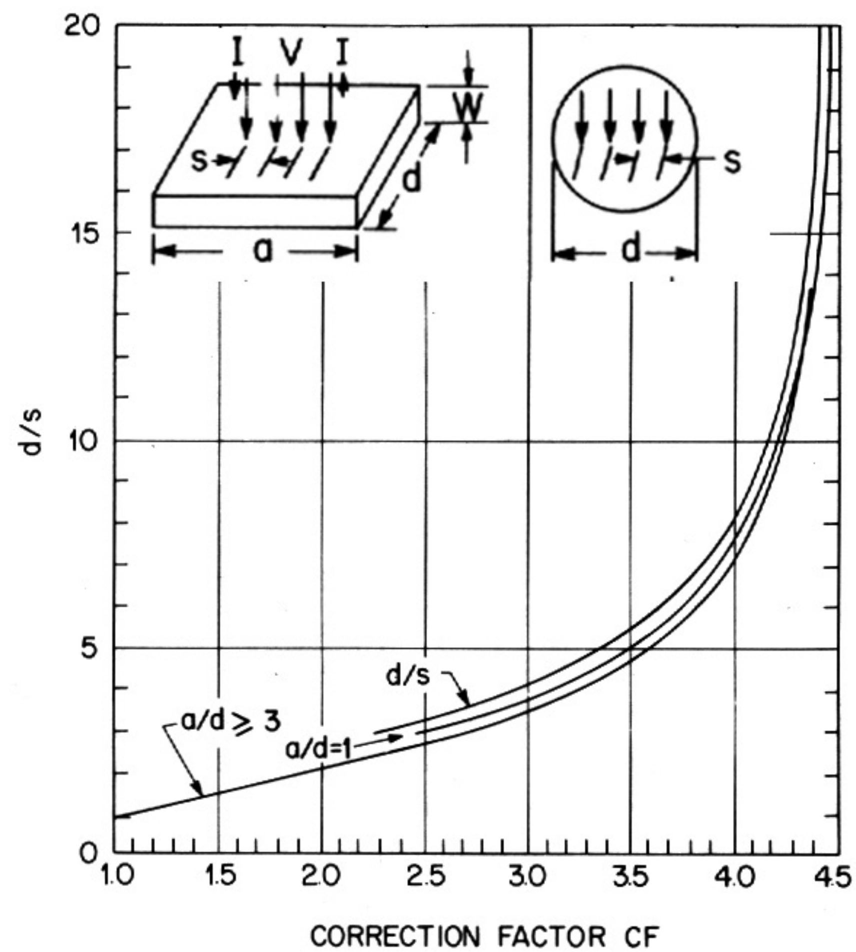
Pierret, Fig. 3.8

Four-Point Probe

- The resistivity of a semiconductor can be determined by a current-voltage measurement using a **four-point probe**

$$\boxed{\rho = \frac{V}{I} W \, CF}$$

- W is the substrate thickness, CF a correction factor (see graph)



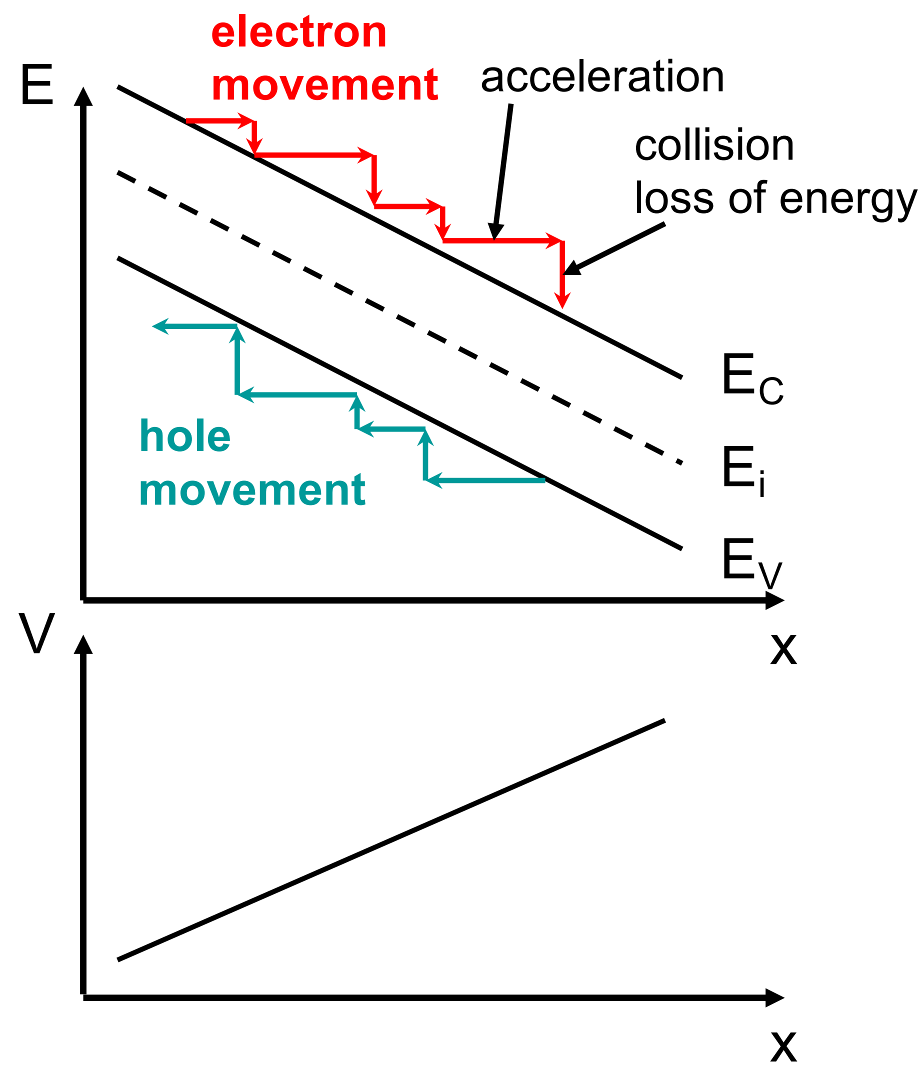
Band Bending by Electric Field

- The presence of an electric field results in a "band bending", i.e. E_C and E_V are no longer constant
- Electrostatic Potential V:

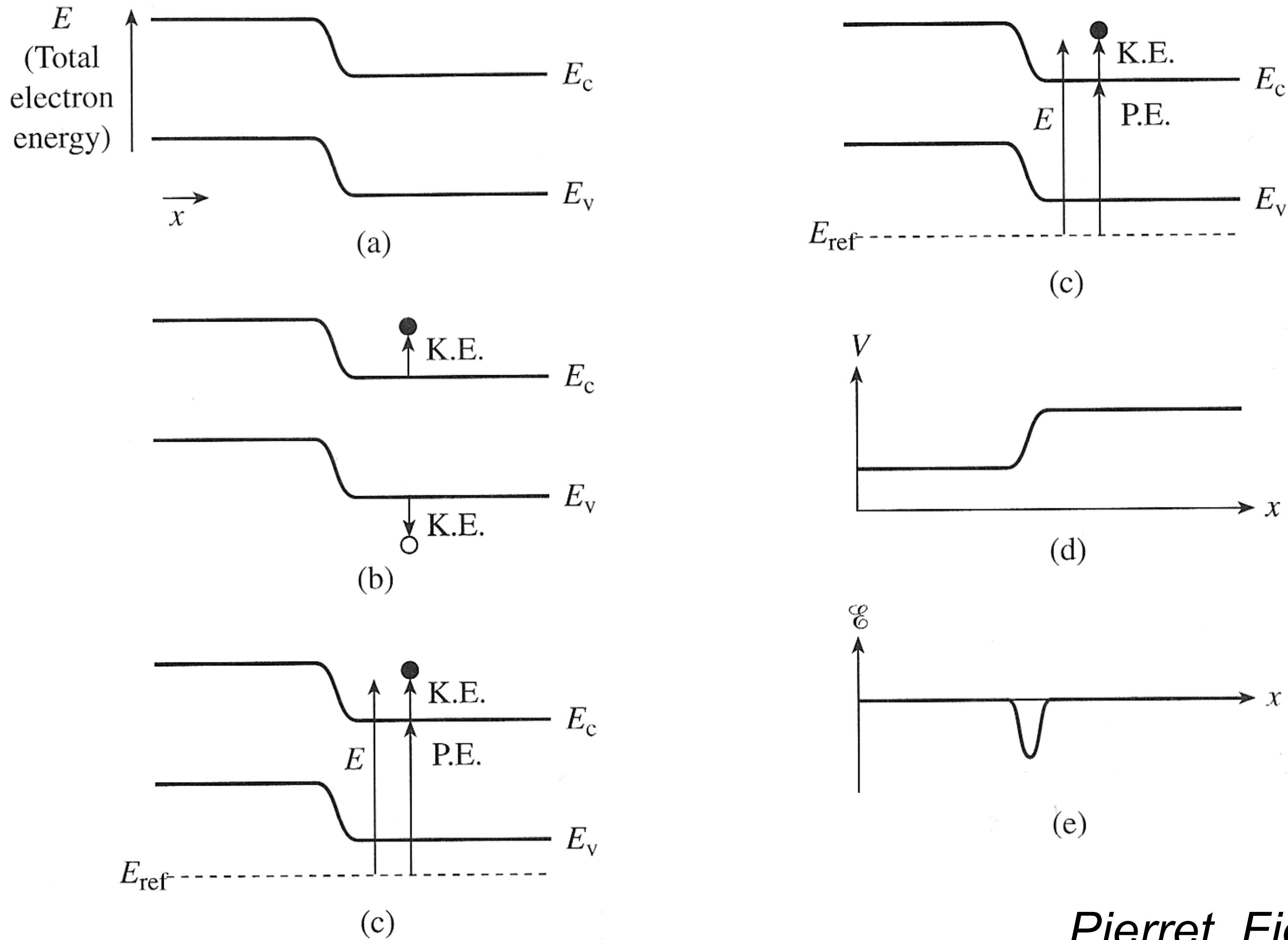
$$\boxed{V = -\frac{1}{q}\left(E_C - E_{ref}\right)}$$

- Note: V and E are arbitrary to within a constant!
- Electrostatic Field ε:

$$\boxed{\begin{aligned} \varepsilon &= -\nabla V \\ \varepsilon &= \frac{1}{q}\frac{dE_C}{dx} = \frac{1}{q}\frac{dE_V}{dx} = \frac{1}{q}\frac{dE_i}{dx} \end{aligned}}$$



Band Bending and Electric Field



Pierret, Fig. 3.10

2.6.2 Carrier Diffusion

- Due to their random thermal motion, particles (in our case electrons and holes) tend to spread out or redistribute, i.e. on a macroscopic scale, they diffuse from areas of higher concentration to areas of lower concentration
- **Diffusion = particle motion due to a gradient in the particle concentration**
- Diffusion is described by **Fick's law**, stating that the particle flux density F [particles/(cm^{2}s)] is proportional to the negative gradient of the particle concentration –grad(η),

$$\boxed{F = -D\,\nabla\eta}$$

with the diffusion coefficient D [cm^2/s] as proportionality constant

Diffusion Current

- In case of diffusion of charged particles, i.e. electron and holes in our case, a diffusion current is resulting (equaling the product of flux density and carrier charge)

$$\boxed{\begin{aligned} J_{p,diff} &= -q\,D_p\,\nabla p \\ J_{n,diff} &= q\,D_n\,\nabla n \end{aligned}}$$

  with diffusion coefficients D_n and D_p [cm^2/s]
- Why has the electron diffusion current no "–" sign?

2.6.3 Total Current Equations

- Summing up drift and diffusion current densities, we obtain the total current density equations:

$$\boxed{\begin{aligned} \vec{J}_p &= q\,\mu_p\, p\,\vec{\varepsilon} - q\,D_p\,\vec{\nabla} p \\ \vec{J}_n &= q\,\mu_n\, n\,\vec{\varepsilon} + q\,D_n\,\vec{\nabla} n \\ \\ \vec{J} &= \vec{J}_n + \vec{J}_p \end{aligned}}$$

2.6.4 Einstein Relations

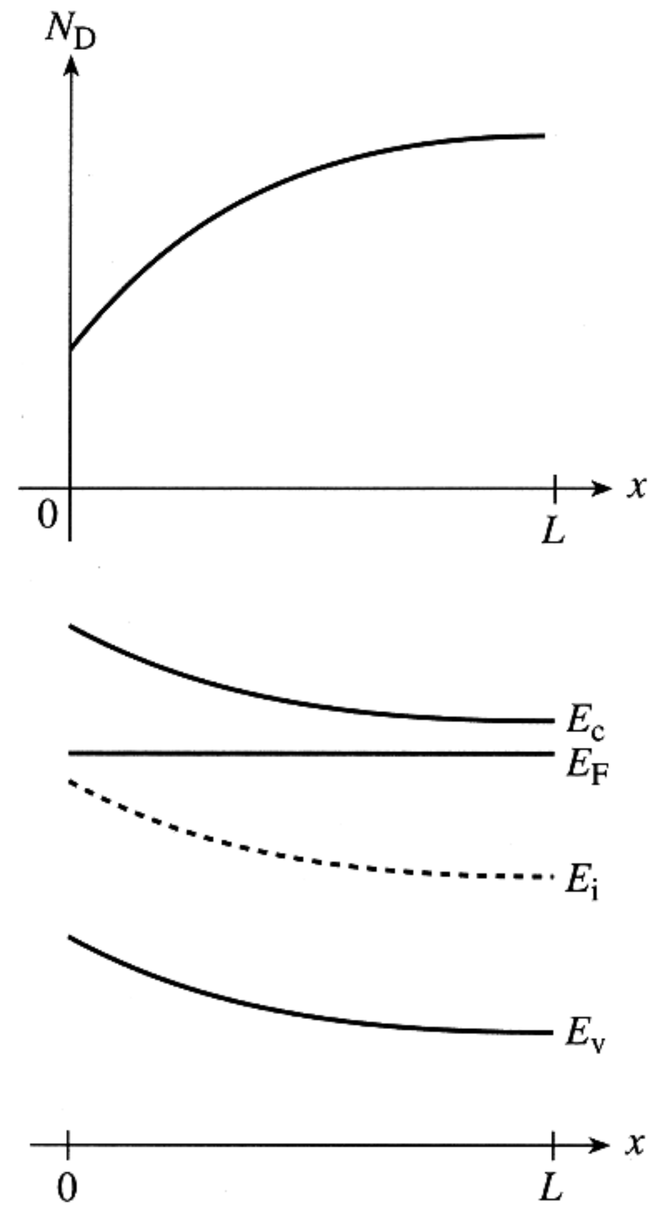
- **Einstein Relations** relate drift (μ) and diffusion (D)

Simplified Derivation:

- Assume non-uniformly doped n-type semiconductor in equilibrium (i.e. E_F = const. across the material)
- In equilibrium, the total electron and hole current densities are zero:

$$\vec{J}_p = q\,\mu_p\,p\,\vec{\varepsilon} - q\,D_p\,\vec{\nabla}p = 0$$

$$\vec{J}_n = q\,\mu_n\,n\,\vec{\varepsilon} + q\,D_n\,\vec{\nabla}n = 0$$



Pierret, Fig. 3.14

Einstein Relations

- From $$n = n_i\, e^{(E_F - E_i)/kT}$$
  we obtain $$\frac{dn}{dx} = -\frac{n_i}{kT} e^{(E_F - E_i)/kT} \frac{dE_i}{dx} = -\frac{q\varepsilon}{kT} \underbrace{n_i e^{(E_F - E_i)/kT}}_{=n}$$
  knowing that $$\varepsilon = \frac{1}{q}\frac{dE_i}{dx} \quad \text{and} \quad \frac{dE_F}{dx} = 0$$

- Inserting into the current density equation for the electrons yields the **Einstein Relations** relating drift (μ_n) and diffusion (D_n)

  (similar derivation for D_p and μ_p)

$$\boxed{D_n = \frac{kT}{q}\mu_n \quad \text{and} \quad D_p = \frac{kT}{q}\mu_p}$$

Drift & Diffusion in Si (T = 300 K)